Trigonometry And Modelling

Chapter Overview

Solution 2

This chapter develops addition and double-angle formulae, combines sine and cosine into one function, and uses these ideas to solve equations and model periodic motion.

1. Addition formulae

$\sin(a+b) = \sin a \cos b + \cos a \sin b$

2. Double-angle formulae

Solve $2\tan 2y \tan y = 3$, $0 \le y < 2\pi$.

3. Combining sine and cosine

Find the maximum of $2\sin x + \cos x$ and where it occurs.

4. Modelling

A tide can be modelled by $x = R\sin(2\pi t/5 + \alpha)$, where t is hours after midnight.

Addition Formulae

Solution 3

Addition formulae handle sums and differences of angles.

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Do I need to memorise these?

They are in the formula booklet, but aim to memorise them, especially sine and cosine.

How to remember

The first function on the right matches the left. Sine mixes sine and cosine; cosine keeps matching pairs. Sine keeps the sign, cosine changes it. Tangent keeps the sign on top and changes it underneath.

The identities apply wherever both sides are defined.

A Common Error

Solution 4

Why is $\sin(A+B)$ not generally $\sin A + \sin B$? Give a counterexample.

Sine is a function, so it need not distribute over addition; similarly $(a+b)^2$ is not generally $a^2 + b^2$.

Take $A = B = 30°$: $\sin 60° = \sqrt{3}/2$, whereas $\sin 30° + \sin 30° = 1$.

Recall the Addition Formulae

Solution 5

Complete $\sin(A+B)$.

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

Complete $\sin(A-B)$.

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

Complete $\cos(A+B)$.

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

Complete $\cos(A-B)$.

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

Complete $\tan(A+B)$.

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

Complete $\tan(A-B)$.

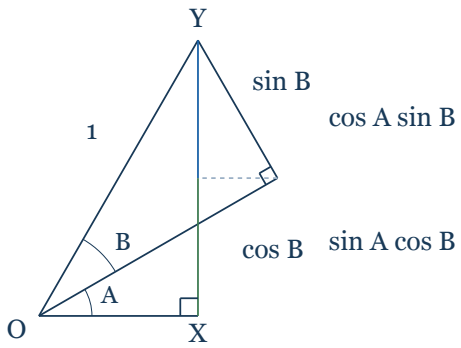
$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

How to remember: the first function on the right matches the left. Sine mixes the functions and keeps the sign; cosine pairs matching functions and changes the sign. Tangent keeps the sign above and changes it below.

Geometric Proof of the Sine Addition Formula

Solution 6

Geometric proof (not required for the exam). Let $OY = 1$ make angle $A + B$ with the horizontal, so $XY = \sin(A+B)$.



1. Resolve the unit line along directions making angles A and $A + \pi/2$ with the horizontal. The perpendicular components have lengths $\cos B$ and $\sin B$.

2. Their vertical components are $\sin A \cos B$ and $\cos A \sin B$.

3. Add the vertical lengths:
   $\sin(A+B) = \sin A \cos B + \cos A \sin B$.

Proof of other identities — Solution 7

Use the sine addition formula to prove the formula for $\sin(A-B)$.

$$\sin(A-B) = \sin(A+(-B))$$
$$= \sin A\cos(-B) + \cos A\sin(-B)$$
$$= \sin A\cos B - \cos A\sin B.$$

Prove the formula for $\cos(A+B)$. Hint: use the complementary-angle relation.

$$\cos(A+B) = \sin((\pi/2 - A) - B)$$
$$= \sin(\pi/2 - A)\cos B - \cos(\pi/2 - A)\sin B$$
$$= \cos A\cos B - \sin A\sin B.$$

Proof of other identities — Solution 8

Edexcel C3 January 2012, Q8(a).

Use the sine and cosine addition formulae to prove the tangent addition formula.

8(a)

$$\tan(A+B) = \frac{\sin A\cos B + \cos A\sin B}{\cos A\cos B - \sin A\sin B}.$$

Divide every numerator and denominator term by $\cos A\cos B$:

$$= \frac{\frac{\sin A\cos B}{\cos A\cos B} + \frac{\cos A\sin B}{\cos A\cos B}}{1 - \frac{\sin A\sin B}{\cos A\cos B}} = \frac{\tan A + \tan B}{1 - \tan A\tan B}.$$

Work where both sides are defined.

Examples

Solution 9

Given $2\sin(x+y) = 3\cos(x-y)$, express $\tan x$ in terms of $\tan y$.

$2\sin x\cos y + 2\cos x\sin y = 3\cos x\cos y + 3\sin x\sin y.$

Where $\cos x\cos y \neq 0$, divide through:

$2\tan x + 2\tan y = 3 + 3\tan x\tan y.$

$\tan x(2 - 3\tan y) = 3 - 2\tan y$

$\tan x = \dfrac{3 - 2\tan y}{2 - 3\tan y}.$

Exercises 7A – Pure Y2 P169

Solution 10

2. Use the sine and cosine subtraction formulae to prove the tangent subtraction formula.

2

$\tan(A-B) = \dfrac{\sin A\cos B - \cos A\sin B}{\cos A\cos B + \sin A\sin B}$

Divide each term by $\cos A\cos B$, where defined:

$= \dfrac{\tan A - \tan B}{1 + \tan A\tan B}.$

3. Substitute $A = P, B = -Q$ into the sine addition formula to obtain $\sin(P-Q)$.

3

$\sin(P-Q) = \sin P\cos(-Q) + \cos P\sin(-Q)$

$= \sin P\cos Q - \cos P\sin Q.$

7. Write $\sin(x+\pi/6) = p\sin x + q\cos x$. Find p, q.

7

$\sin(x+\pi/6) = \dfrac{\sqrt{3}}{2}\sin x + \dfrac{1}{2}\cos x.$

$p = \sqrt{3}/2,\ q = 1/2.$

Exercises 7A – Pure Y2 P169

Solution 11

11. Given $\cos y = \sin(x + y)$, show that $\tan y = \sec x - \tan x$.

11

$\cos y = \sin x \cos y + \cos x \sin y.$

Divide by $\cos x \cos y$, where defined:

$\sec x = \tan x + \tan y \quad \Rightarrow \quad \tan y = \sec x - \tan x.$

15. Given $\tan(x + \pi/3) = 1/2$, show that $\tan x = 8 - 5\sqrt{3}$.

15

Let $t = \tan x$. Then

$$\frac{t + \sqrt{3}}{1 - \sqrt{3}t} = \frac{1}{2} \Rightarrow (2 + \sqrt{3})t = 1 - 2\sqrt{3}.$$

$$t = \frac{(1 - 2\sqrt{3})(2 - \sqrt{3})}{(2 + \sqrt{3})(2 - \sqrt{3})} = 8 - 5\sqrt{3}.$$

Exercises 7A – Pure Y2 P169

Solution 12

16. Prove $\cos\theta + \cos(\theta + 2\pi/3) + \cos(\theta + 4\pi/3) = 0$. Show every stage of working.

16

Put $\phi = \theta + 2\pi/3$. Then

$\cos(\phi - 2\pi/3) = \cos\phi\cos(2\pi/3) + \sin\phi\sin(2\pi/3),$

$\cos(\phi + 2\pi/3) = \cos\phi\cos(2\pi/3) - \sin\phi\sin(2\pi/3).$

Adding these and $\cos\phi$, the sine terms cancel:

$2\cos\phi\cos(2\pi/3) + \cos\phi = 2\cos\phi(-1/2) + \cos\phi = 0.$

Uses of Addition Formulae

Solution 13

Using a suitable angle formula, show $\sin 15^\circ = (\sqrt{6} - \sqrt{2})/4$.

$$\sin 15^\circ = \sin(45^\circ - 30^\circ)$$
$$= \frac{1}{\sqrt{2}}\frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}\frac{1}{2} = \frac{\sqrt{6}-\sqrt{2}}{4}.$$

Given $\sin A = -3/5$, $180^\circ < A < 270^\circ$, and $\cos B = -12/13$ with B obtuse, find (a) $\cos(A - B)$, (b) $\tan(A + B)$.

Tip: $\cos A = \pm\sqrt{1 - \sin^2 A}$. Choose the sign from the quadrant.

a

$\cos A = -4/5$ and $\sin B = 5/13$.

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$
$$= (-4/5)(-12/13) + (-3/5)(5/13) = 33/65.$$

Part (b) continues on the next slide.

Continued...

Solution 14

Given that $\sin A = -\frac{3}{5}$ and $180^\circ < A < 270^\circ$, and that $\cos B = -\frac{12}{13}$, and that B is obtuse, find the value of:
(b) $\tan(A + B)$

b

From the previous slide,

$\sin A = -3/5, \cos A = -4/5,$

$\sin B = 5/13, \cos B = -12/13.$

$\tan A = 3/4, \qquad \tan B = -5/12.$

$$\tan(A + B) = \frac{3/4 - 5/12}{1 - (3/4)(-5/12)} = \frac{1/3}{21/16} = \frac{16}{63}.$$

Test Your Understanding

Solution 15

Without a calculator, find $\cos 75°$.

a

$$\cos(45° + 30°) = \frac{1}{\sqrt{2}}\frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}\frac{1}{2} = \frac{\sqrt{6} - \sqrt{2}}{4}.$$

Without a calculator, find $\tan 75°$.

b

$$\sin 75° = \sin 45° \cos 30° + \cos 45° \sin 30°$$

$$= \frac{1}{\sqrt{2}}\frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}}\frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

$$\tan 75° = \frac{\sqrt{6} + \sqrt{2}}{\sqrt{6} - \sqrt{2}}$$

$$= \frac{(\sqrt{6} + \sqrt{2})^2}{6 - 2} = \frac{8 + 4\sqrt{3}}{4} = 2 + \sqrt{3}.$$

Challenging question

Solution 16

Without a calculator. Edexcel June 2013 Q3.

a. Given $2\cos(x + 50°) = \sin(x + 40°)$, prove $\tan x = \frac{1}{3}\tan 40°$.

a

Expand both sides, using $\cos 50° = \sin 40°$ and $\sin 50° = \cos 40°$:

$2\cos x \sin 40° - 2\sin x \cos 40°$

$= \sin x \cos 40° + \cos x \sin 40°.$

$\cos x \sin 40° = 3\sin x \cos 40°.$

Divide by $3\cos x \cos 40°$:

$\tan x = \frac{1}{3}\tan 40°.$

b. Solve $2\cos(2\theta + 50°) = \sin(2\theta + 40°)$, $0 \leq \theta < 360°$.

b

$\tan 2\theta = \frac{1}{3}\tan 40°.$

$2\theta = 15.626\ldots° + 180°k.$

In $0 \leq 2\theta < 720°$:

$\theta = 7.8°, 97.8°, 187.8°, 277.8°$ (1 d.p.).

Exercise 7B - Pure Y2 P172-173

Solution 17

3. a. Use the tangent addition formula for $\tan(45° + 30°)$.

b. Hence show $\tan 75° = 2 + \sqrt{3}$.

3a

$$\tan(45° + 30°) = \frac{\tan 45° + \tan 30°}{1 - \tan 45° \tan 30°} = \frac{\sqrt{3}+1}{\sqrt{3}-1}.$$

3b

$$\frac{(\sqrt{3}+1)^2}{3-1} = \frac{4+2\sqrt{3}}{2} = 2 + \sqrt{3}.$$

5. a. Show $\cos 105° = (\sqrt{2} - \sqrt{6})/4$.

b. Write $\sec 105° = -\sqrt{a}(1 + \sqrt{b})$, finding integers a, b.

5a

$$\cos 105° = \cos 60° \cos 45° - \sin 60° \sin 45°$$
$$= \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4} = \frac{\sqrt{2}-\sqrt{6}}{4}.$$

5b

$$\sec 105° = \frac{4}{\sqrt{2}-\sqrt{6}} = -(\sqrt{2}+\sqrt{6})$$
$$= -\sqrt{2}(1+\sqrt{3}).$$
$$a = 2, b = 3.$$

Exercise 7B - Pure Y2 P172-173

Solution 18

8. $\sin A = 8/17$, A acute; $\cos B = -4/5$, B obtuse. Find (a) $\sin(A - B)$, (b) $\cos(A - B)$, (c) $\cot(A - B)$.

General observations: use the quadrants to select the signs of the missing ratios.

a

$\cos A = 15/17$, $\sin B = 3/5$.

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$
$$= \frac{8}{17}\left(-\frac{4}{5}\right) - \frac{15}{17}\frac{3}{5} = -\frac{77}{85}.$$

b

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$
$$= \frac{15}{17}\left(-\frac{4}{5}\right) + \frac{8}{17}\frac{3}{5} = -\frac{36}{85}.$$

c

$\tan A = 8/15$, $\tan B = -3/4$.

$$\tan(A - B) = \frac{8/15 + 3/4}{1 - (8/15)(3/4)} = \frac{77/60}{3/5} = 77/36.$$

Hence $\cot(A - B) = 36/77$. This also equals $(-36/85)/(-77/85)$.

Exercise 7B - Pure Y2 P172-173

Solution 19

Without a calculator. 10. $\tan A = 1/5$, $\tan B = 2/3$. Find $A + B$ when (a) both are acute, (b) A is reflex and B acute.

a

$$\tan(A+B) = \frac{1/5 + 2/3}{1 - (1/5)(2/3)} = 1.$$

If both are acute, $0 < A + B < 180°$, so $A + B = 45°$.

b

A reflex with positive tangent lies in quadrant 3. Increasing the acute value of A by $180°$ gives $A + B = 225°$.

Extension: Difference Formulae

Solution 20

STEP I 2010 Q3.

a. Show

$$\sin(x+y) - \sin(x-y) = 2\cos x \sin y.$$

Hence show

$$\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}.$$

a

$\sin(x+y) - \sin(x-y)$

$= \sin x \cos y + \cos x \sin y - (\sin x \cos y - \cos x \sin y)$

$= 2\cos x \sin y.$

Put $x = (A+B)/2, y = (A-B)/2$ to obtain the required difference of sines.

b. Show

$$\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}.$$

b

$\cos(x+y) - \cos(x-y)$

$= \cos x \cos y - \sin x \sin y - (\cos x \cos y + \sin x \sin y)$

$= -2\sin x \sin y.$

Use the same substitutions for x, y.

c. Points P, Q, R, S have coordinates $(a\cos u, b\sin u)$ at $u = p, q, r, s$, with $a, b > 0$ and $0 \le p < q < r < s < 2\pi$. Assuming PQ, SR are not vertical, prove they are parallel exactly when $r + s - p - q = 2\pi$.

c

$$m_{PQ} = \frac{b(\sin q - \sin p)}{a(\cos q - \cos p)}$$

$$= \frac{2b\cos((p+q)/2)\sin((q-p)/2)}{-2a\sin((p+q)/2)\sin((q-p)/2)}$$

$$= -\frac{b}{a}\cot\frac{p+q}{2}.$$

Similarly $m_{SR} = -(b/a)\cot((r+s)/2)$. Equal slopes occur iff $(r+s-p-q)/2 = k\pi$. Since $0 < r + s - p - q < 4\pi$, only $k = 1$ is possible: the difference is 2π. Conversely this difference gives equal slopes.

Double Angle Formulae

Solution 21

$\sin 2A = 2\sin A\cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$= 2\cos^2 A - 1 = 1 - 2\sin^2 A$

$\tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$

Tip: in the cosine identities, the cosine term is positive and the sine term negative. The first cosine form is used less often.

Set $B = A$ in the addition formulae. For example:

$\sin 2A = \sin(A + A)$

$= \sin A\cos A + \cos A\sin A = 2\sin A\cos A.$

Double-angle formulae relate an angle to half that angle. Use them in either direction, where defined.

Examples

Solution 22

a. Write $\cos^2 50° - \sin^2 50°$ as one ratio.

b. Simplify $\dfrac{2\tan(\pi/6)}{1-\tan^2(\pi/6)}$.

c. Simplify $\dfrac{4\sin 70°}{\sec 70°}$.

a

Matches $\cos 2A = \cos^2 A - \sin^2 A$:

$\cos^2 50° - \sin^2 50° = \cos 100°.$

b

Matches the tangent double-angle formula:

$\dfrac{2\tan(\pi/6)}{1-\tan^2(\pi/6)} = \tan(\pi/3).$

c

$4\sin 70°\cos 70° = 2\sin 140°.$

Examples

Solution 23

Given $x = 3\sin\theta$ and $y = 3 - 4\cos 2\theta$, eliminate θ.

$$y = 3 - 4(1 - 2\sin^2\theta) = 8\sin^2\theta - 1.$$
$$\sin\theta = x/3 \Rightarrow y = 8x^2/9 - 1, \quad -3 \le x \le 3.$$

This changes parametric equations into a Cartesian equation, as in the next chapter.

Given $\cos x = 3/4$ and x acute, find (a) $\sin 2x$, (b) $\tan 2x$.

a

$\sin x = \sqrt{(1 - 9/16)} = \sqrt{7}/4$,
positive in quadrant 1.
$\sin 2x = 2(\sqrt{7}/4)(3/4) = 3\sqrt{7}/8$.

b

$$\tan x = (\sqrt{7}/4)/(3/4) = \sqrt{7}/3.$$
$$\tan 2x = \frac{2\sqrt{7}/3}{1 - 7/9} = 3\sqrt{7}.$$

Exercise 7C – Pure Y2 p175-177

Solution 24

3. Use $\tan(A + A)$ to express $\tan 2A$ in terms of $\tan A$.

3

$$\tan 2A = \frac{\tan A + \tan A}{1 - \tan A \tan A} = \frac{2\tan A}{1 - \tan^2 A}.$$

6. a. Show $(\sin A + \cos A)^2 = 1 + \sin 2A$.
b. Hence find $(\sin(\pi/8) + \cos(\pi/8))^2$.

6a

$$\sin^2 A + \cos^2 A + 2\sin A\cos A$$
$$= 1 + \sin 2A.$$

6b

$$1 + \sin(\pi/4) = 1 + \sqrt{2}/2 = \frac{2 + \sqrt{2}}{2}.$$

8. $p = 2\cos\theta, q = \cos 2\theta$. Express q in terms of p.

8

$$q = 2\cos^2\theta - 1 = 2(p/2)^2 - 1 = p^2/2 - 1.$$

Exercise 7C – Pure Y2 p175-177 Solution 25

11. Find the possible values of $\sin\theta$ if $\cos 2\theta = 23/25$.

11

$23/25 = 1 - 2\sin^2\theta \Rightarrow \sin^2\theta = 1/25.$

$\sin\theta = \pm 1/5.$

14. Given $\pi < \theta < 3\pi/2$ and $\tan\theta = 3/4$, find $\tan(\theta/2)$.

14

Let $t = \tan(\theta/2)$.

$$\frac{3}{4} = \frac{2t}{1-t^2} \Rightarrow 3t^2 + 8t - 3 = 0$$

$\Rightarrow (3t-1)(t+3) = 0.$

Since $\pi/2 < \theta/2 < 3\pi/4$, tangent is negative:

$t = -3.$

Exercise 7C – Pure Y2 p175-177 Solution 26

15. Given $\cos x + \sin x = m$, $\cos x - \sin x = n$, express $\cos 2x$ in terms of m, n.

15

Multiply the equations:

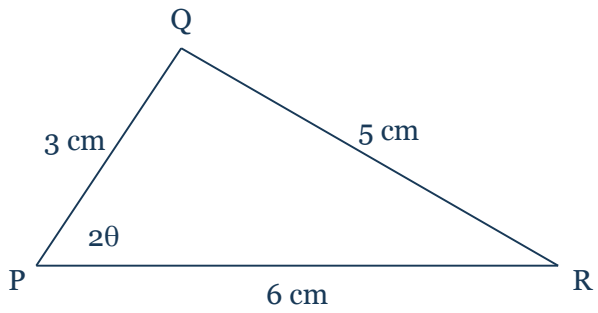
$(\cos x + \sin x)(\cos x - \sin x) = mn.$

$\cos 2x = \cos^2 x - \sin^2 x = mn.$

16. In triangle PQR, $PQ = 3$, $PR = 6$, $QR = 5$ cm and $\angle QPR = 2\theta$.

a. Use the cosine rule to show $\cos 2\theta = 5/9$.

b. Find the exact value of $\sin\theta$.



16a

$$\cos 2\theta = \frac{3^2 + 6^2 - 5^2}{2(3)(6)} = 5/9.$$

16b

$5/9 = 1 - 2\sin^2\theta \Rightarrow \sin^2\theta = 2/9.$

Since $0 < \theta < \pi/2$, $\sin\theta = \sqrt{2}/3.$

Solving Trigonometric Equations

Solution 27

This is effectively the same type of question you encountered in Chapter 6 and in Year 1, except you may need to use either the addition formulae or double angle formulae.

[Textbook] Solve $3\cos 2x - \cos x + 2 = 0$ for $0 \leq x \leq 360°$.

Use $\cos 2x = 2\cos^2 x - 1$ so the equation contains only cosine:

$3(2\cos^2 x - 1) - \cos x + 2 = 0$

$6\cos^2 x - \cos x - 1 = 0$

$(3\cos x + 1)(2\cos x - 1) = 0.$

$\cos x = -1/3 \quad \text{or} \quad \cos x = 1/2.$

$x = 60°, 109.5°, 250.5°, 300°.$

Further Examples

Solution 28

a. Show $\sin 3A = 3\sin A - 4\sin^3 A$.

b. Hence solve $16\sin^3\theta - 12\sin\theta - 2\sqrt{3} = 0$, $0 < \theta < 2\pi$.

a

$\sin(2A + A) = \sin 2A\cos A + \cos 2A\sin A$

$= 2\sin A\cos^2 A + (1 - 2\sin^2 A)\sin A$

$= 2\sin A(1 - \sin^2 A) + \sin A - 2\sin^3 A$

$= 3\sin A - 4\sin^3 A.$

b

$-4\sin 3\theta = 2\sqrt{3} \Rightarrow \sin 3\theta = -\sqrt{3}/2.$

Since $0 < 3\theta < 6\pi$:

$3\theta = 4\pi/3, 5\pi/3, 10\pi/3, 11\pi/3, 16\pi/3, 17\pi/3.$

$\theta = 4\pi/9, 5\pi/9, 10\pi/9, 11\pi/9, 16\pi/9, 17\pi/9.$

Solve $4\cos(\theta - 30°) = 8\sqrt{2}\sin\theta$, $0 \leq \theta < 360°$.

Expand first:

$4\cos\theta\cos 30° + 4\sin\theta\sin 30° = 8\sqrt{2}\sin\theta.$

$2\sqrt{3}\cos\theta + 2\sin\theta = 8\sqrt{2}\sin\theta.$

$\tan\theta = \dfrac{2\sqrt{3}}{8\sqrt{2} - 2}.$

$\theta = 20.4°, 200.4°.$

Test Your Understanding — Solution 29

Edexcel C3 January 2013 Q6.

i. Without a calculator, find exactly $(\sin 22.5^\circ + \cos 22.5^\circ)^2$, showing working.

ii. Show $\cos 2\theta + \sin\theta = 1$ becomes $k\sin^2\theta - \sin\theta = 0$, stating k. Hence solve for $0 \le \theta < 360^\circ$.

If you finish: solve $2\tan 2y \tan y = 3$, $0 \le y < 2\pi$, to 2 d.p.

Tip: remember both signs when taking a square root.

i

$$(\sin 22.5^\circ + \cos 22.5^\circ)^2$$
$$= \sin^2 22.5^\circ + 2\sin 22.5^\circ \cos 22.5^\circ + \cos^2 22.5^\circ$$
$$= 1 + \sin 45^\circ = 1 + \sqrt{2}/2.$$

ii

$$1 - 2\sin^2\theta + \sin\theta = 1 \Rightarrow \sin\theta(2\sin\theta - 1) = 0.$$
$$k = 2;\ \theta = 0^\circ, 30^\circ, 150^\circ, 180^\circ.$$

Extra

$$\frac{4\tan^2 y}{1 - \tan^2 y} = 3 \Rightarrow 7\tan^2 y = 3 \Rightarrow \tan y = \pm\sqrt{3/7}.$$
$$y = 0.58, 2.56, 3.72, 5.70.$$
All lie in the original domain.

Exercise 7D – Pure Y2 P180-181 — Solution 30

2. a. Show $\sin(\theta + \pi/4) = (\sin\theta + \cos\theta)/\sqrt{2}$.

b. Hence solve $(\sin\theta + \cos\theta)/\sqrt{2} = 1/\sqrt{2}$, $0 \le \theta \le 2\pi$.

c. Deduce the solutions of $\sin\theta + \cos\theta = 1$ in this interval.

a

$$\sin(\theta + \pi/4) = \sin\theta\frac{1}{\sqrt{2}} + \cos\theta\frac{1}{\sqrt{2}}.$$

b

$$\pi/4 \le \theta + \pi/4 \le 9\pi/4.$$
So $\theta + \pi/4 = \pi/4, 3\pi/4, 9\pi/4$, giving
$\theta = 0, \pi/2, 2\pi$.

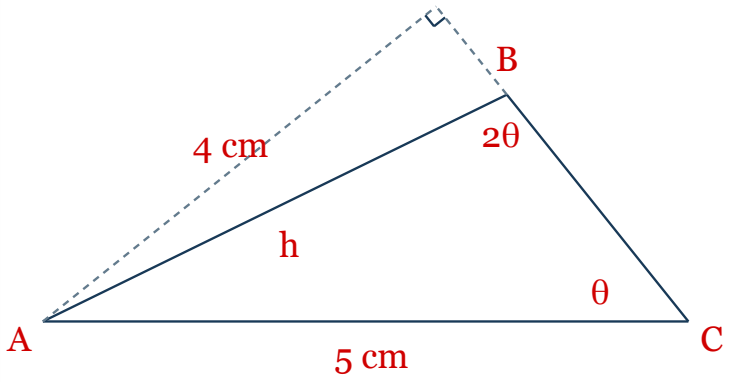
c

The equation in (c) is equivalent to (b), so again
$\theta = 0, \pi/2, 2\pi$.

Exercise 7D – Pure Y2 P180-181 Solution 31

6. In triangle ABC, $AB = 4$ cm, $AC = 5$ cm, $\angle ABC = 2\theta$, $\angle ACB = \theta$. Find θ to 1 d.p.

6 diagram



6

The altitude from A to BC is $h = 4\sin 2\theta = 5\sin\theta$. Equivalently, the sine rule gives $4/\sin\theta = 5/\sin 2\theta$.

$\sin\theta(8\cos\theta - 5) = 0$.

The angles give $0 < \theta < 60°$, so $\sin\theta \neq 0$.

$\theta = \cos^{-1}(5/8) = 51.3°$.

7. a. Write $5\sin 2\theta + 4\sin\theta = 0$ as $a\sin\theta(b\cos\theta + c) = 0$, stating a, b, c.

b. Solve for $0 \leq \theta < 360°$.

7a

$10\sin\theta\cos\theta + 4\sin\theta = 2\sin\theta(5\cos\theta + 2)$.

$a = 2, b = 5, c = 2$.

7b

$\sin\theta = 0$ or $\cos\theta = -2/5$.

$\theta = 0°, 113.6°, 180°, 246.4°$.

Exercise 7D – Pure Y2 P180-181 Solution 32

8. a. Show $\sin 2\theta + \cos 2\theta = 1$ gives $2\sin\theta(\cos\theta - \sin\theta) = 0$.

b. Solve for $0 \leq \theta < 360°$.

8a

$2\sin\theta\cos\theta + 1 - 2\sin^2\theta = 1$

$\Rightarrow 2\sin\theta(\cos\theta - \sin\theta) = 0$.

8b

$\sin\theta = 0$ or $\cos\theta = \sin\theta$.

Hence $\theta = 0°, 45°, 180°, 225°$.

9a. Prove $(\cos 2\theta - \sin 2\theta)^2 = 1 - \sin 4\theta$.

9a

$\cos^2 2\theta - 2\sin 2\theta\cos 2\theta + \sin^2 2\theta$

$= 1 - \sin 4\theta$.

Exercise 7D – Pure Y2 P180-181 — Solution 33

11a. Show $3\cos^2 x - \sin^2 x = 1 + 2\cos 2x$.

a

$1 + 2\cos 2x = (\cos^2 x + \sin^2 x) + 2(\cos^2 x - \sin^2 x)$
$= 3\cos^2 x - \sin^2 x.$

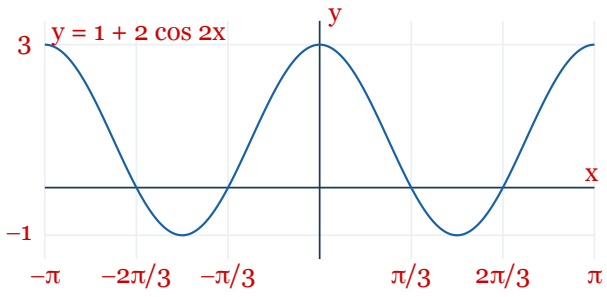
b. Sketch $y = 3\cos^2 x - \sin^2 x$, $-\pi \le x \le \pi$, labelling intercepts.

b working

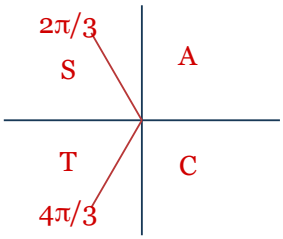
Compress $\cos x$ horizontally by factor $1/2$, stretch vertically by 2, then translate up 1.

At $x = 0$, $y = 3$. For $y = 0$, $\cos 2x = -1/2$, so $x = \pm\pi/3, \pm 2\pi/3$.

b sketch



b reference angles



Extension — Solution 34

1a. Use $\sin(A - B)$ to show $\sin(90° - x) = \cos x$.

1b. Solve $2\sin(\theta + 17°) = \cos(\theta + 8°)/\cos(\theta + 17°)$, $0 < \theta < 360°$.

2a. Solve $\sin(\pi/3 - \theta) = \cos\theta/\sqrt{3}$, $0 \le \theta < 2\pi$.

2b. Find x when $\arcsin(1 - 2x) = \pi/3 - \arcsin x$, $0 < x < 0.5$.

Solutions to question 2 follow on the next slide.

1a

$\sin 90° \cos x - \cos 90° \sin x = \cos x.$

1b

$\sin(2\theta + 34°) = \cos(\theta + 8°) = \sin(82° - \theta).$

Either $2\theta + 34 = 82 - \theta + 360k$, giving $\theta = 16 + 120k$, or $2\theta + 34 = 98 + \theta + 360k$, giving $\theta = 64 + 360k$.

Thus $\theta = 16°, 64°, 136°, 256°$; none makes the original denominator zero.

Solution to Extension Question 2

2a. Solve $\sin(\pi/3 - \theta) = \cos\theta/\sqrt{3}$, $0 \le \theta < 2\pi$.

2a

$$\frac{\sqrt{3}}{2}\cos\theta - \frac{1}{2}\sin\theta = \frac{1}{\sqrt{3}}\cos\theta$$

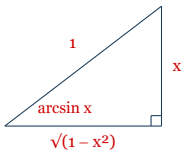
$$\Rightarrow \sin\theta = \frac{1}{\sqrt{3}}\cos\theta.$$

Here $\cos\theta \neq 0$, so $\tan\theta = 1/\sqrt{3}$:

$\theta = \pi/6, 7\pi/6$.

2b. Solve $\arcsin(1 - 2x) = \pi/3 - \arcsin x$, $0 < x < 0.5$.

2b triangle



2b

Take sine and use the principal range:

$$1 - 2x = \frac{\sqrt{3}}{2}\sqrt{1 - x^2} - \frac{x}{2}.$$

$$2 - 3x = \sqrt{3}\sqrt{1 - x^2}.$$

Both sides are positive in this interval; square:

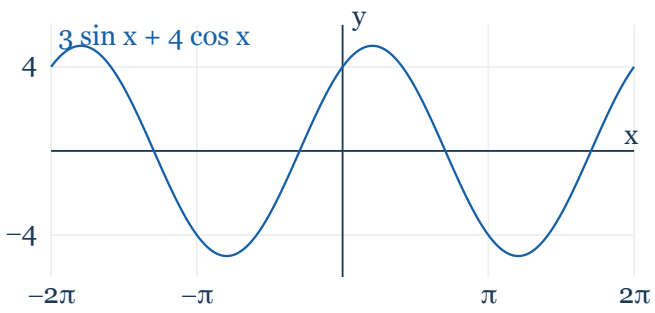
$$4 - 12x + 9x^2 = 3 - 3x^2 \Rightarrow 12x^2 - 12x + 1 = 0.$$

$$x = \frac{3 \pm \sqrt{6}}{6}.$$

Only $x = (3 - \sqrt{6})/6$ lies in the interval. Both original angles are in $(0, \pi/2)$, so equality of their sines verifies the original equation.

Combining Sine and Cosine

Sketch $y = 3\sin x + 4\cos x$. What do you notice?



It is a sine curve stretched vertically and translated horizontally.

Write $y = R\sin(x + \alpha)$: the amplitude is R, and the horizontal translation is α to the left.

Combining Sine and Cosine

Solution 37

Write $3\sin x + 4\cos x$ as $R\sin(x+\alpha)$, giving α in degrees to 1 d.p.

To find R, square the two coefficient equations and add:

$R^2(\cos^2\alpha + \sin^2\alpha) = 3^2 + 4^2$.

1. Expand

Expand: $R\sin x\cos\alpha + R\cos x\sin\alpha$.

2. Compare

Compare coefficients: $R\cos\alpha = 3,\ R\sin\alpha = 4$.

3. Amplitude

$R = \sqrt{3^2+4^2} = 5$.

4. Phase

$\tan\alpha = 4/3$, with α acute, so $\alpha = 53.1^\circ$ to 1 d.p.

5. Result

$3\sin x + 4\cos x = 5\sin(x+\alpha), \quad \alpha = \tan^{-1}(4/3)$.

Thus approximately $5\sin(x+53.1^\circ)$.

Test Your Understanding

Solution 38

Write $\sin x + \cos x = R\sin(x+\alpha)$, giving α in terms of π.

Expand:

$R\sin(x+\alpha) = R\sin x\cos\alpha + R\cos x\sin\alpha$.

$R\cos\alpha = 1, R\sin\alpha = 1$.

Thus $R = \sqrt{2}, \tan\alpha = 1, \alpha = \pi/4$.

$\sin x + \cos x = \sqrt{2}\sin(x+\pi/4)$.

Write $\sin x - \sqrt{3}\cos x = R\sin(x-\alpha)$, giving α in terms of π.

Expand:

$R\sin(x-\alpha) = R\sin x\cos\alpha - R\cos x\sin\alpha$.

$R\cos\alpha = 1, R\sin\alpha = \sqrt{3}$.

Thus $R = 2, \tan\alpha = \sqrt{3}, \alpha = \pi/3$.

$\sin x - \sqrt{3}\cos x = 2\sin(x-\pi/3)$.

Further Examples

Solution 39

Write $2\cos\theta + 5\sin\theta = R\cos(\theta - \alpha)$, $0 < \alpha < 90°$.
Hence solve $2\cos\theta + 5\sin\theta = 3$, $0 < \theta < 360°$.

$R\cos\alpha = 2$, $R\sin\alpha = 5$, so $R = \sqrt{29}$,
$\alpha = \tan^{-1}(5/2)$.
$\cos(\theta - \alpha) = 3/\sqrt{29}$.
Use the unrounded α and both angle branches:
$\theta = 12.1°, 124.3°$.

Without calculus, find the maximum of $12\cos\theta + 5\sin\theta$, and the smallest positive θ at which it occurs.

$12\cos\theta + 5\sin\theta = 13\cos(\theta - \alpha)$,
$\alpha = \tan^{-1}(5/12)$.
Cosine is at most 1, so the maximum is 13, first at $\theta = \alpha = 22.6°$ (1 d.p.).

Quickfire Maxima

Solution 40

Find the maximum and the smallest positive angle (degrees): $20\sin\theta$.

1 maximum
Maximum 20.

1 angle
First at $\theta = 90°$.

Find the maximum and the smallest positive angle (degrees): $5 - 10\sin\theta$.

2 maximum
Maximum 15.

2 angle
First at $\theta = 270°$.

Find the maximum and the smallest positive angle (degrees): $3\cos(\theta + 20°)$.

3 maximum
Maximum 3.

3 angle
First at $\theta = 340°$.

Find the maximum and the smallest positive angle (degrees): $\frac{2}{10+3\sin(\theta-30°)}$.

4 maximum
Maximum $2/7$.

4 angle
First at $\theta = 300°$.

Further Test Your Understanding

Solution 41

Edexcel C3 January 2013 Q4.

a. Express $6\cos\theta + 8\sin\theta = R\cos(\theta - \alpha)$, $R > 0, 0 < \alpha < \pi/2$, with α to 3 d.p.

b. For $p(\theta) = 4/(12 + 6\cos\theta + 8\sin\theta)$, $0 \leq \theta \leq 2\pi$, find (i) its maximum, (ii) where it occurs.

a $R = \sqrt{6^2 + 8^2} = 10$, $\alpha = \tan^{-1}(8/6) = 0.927$ (3 d.p.).

b(i)

$$p(\theta) = \frac{4}{12 + 10\cos(\theta - \alpha)}.$$

The positive denominator is at least 2, so the maximum is 2.

b(ii)

$\cos(\theta - \alpha) = -1$, so $\theta = \pi + \alpha = 4.069$ radians (3 d.p.).

Exercise 7E – Pure Y2 P184-186

Solution 42

6. $f(\theta) = \sin\theta + 3\cos\theta = R\sin(\theta + \alpha)$, $R > 0, 0 < \alpha < 90°$.

a. Find R, α.

b. Solve $f(\theta) = 2$, $0 \leq \theta < 360°$.

a

Comparing coefficients: $R\cos\alpha = 1, R\sin\alpha = 3$.

Thus $R = \sqrt{10}, \alpha = \tan^{-1} 3 = 71.6°$ (1 d.p.).

b

Use unrounded α:

$\sin(\theta + \alpha) = 2/\sqrt{10}$.

The reference angle is $39.2315\ldots°$. Since $\alpha \leq \theta + \alpha < 360° + \alpha$, choose $180° - 39.2315\ldots°$ and $360° + 39.2315\ldots°$.

$\theta = 69.2°, 327.7°$.

Exercise 7E – Pure Y2 P184-186 — Solution 43

12. a. Given $\cot\theta + 2 = \operatorname{cosec}\theta$, show $2\sin\theta + \cos\theta = 1$.

b. Solve the original equation, $0 \le \theta < 360°$.

a

$\dfrac{\cos\theta}{\sin\theta} + 2 = \dfrac{1}{\sin\theta}$.

Multiply by $\sin\theta \neq 0$: $\cos\theta + 2\sin\theta = 1$.

b

Write $R\sin(\theta+\alpha)$. Comparing coefficients, $R\cos\alpha = 2, R\sin\alpha = 1$; hence $R = \sqrt{5}, \alpha = \tan^{-1}(1/2)$.

$\sin(\theta+\alpha) = 1/\sqrt{5}$.

In $\alpha \le \theta + \alpha < 360° + \alpha$, the shifted angles are α and $180° - \alpha$, giving candidates $0°, 126.8699\ldots°$.

Reject $0°$, where cotangent and cosecant are undefined. Thus $\theta = 126.9°$.

Proving Trigonometric Identities — Solution 44

Just like Chapter 6 had 'provey' and 'solvey' questions, we also get the 'provey' questions in Chapter 7. Just use the appropriate double angle or addition formula.

Prove $\tan 2\theta = 2/(\cot\theta - \tan\theta)$.

$\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$

$= \dfrac{2}{1/\tan\theta - \tan\theta} = \dfrac{2}{\cot\theta - \tan\theta}$.

The identity applies on the common domain.

Prove $(1-\cos 2\theta)/\sin 2\theta = \tan\theta$.

$\dfrac{1-(1-2\sin^2\theta)}{2\sin\theta\cos\theta} = \dfrac{2\sin^2\theta}{2\sin\theta\cos\theta} = \tan\theta$.

Work where the original denominator is nonzero.

Test Your Understanding — Solution 45

[OCR] Prove $\cot 2x + \operatorname{cosec} 2x = \cot x$.

$$\frac{\cos 2x + 1}{\sin 2x} = \frac{2\cos^2 x}{2\sin x \cos x} = \cot x.$$

Valid wherever the left side is defined.

[OCR] By writing $\cos x = \cos(2 \times x/2)$, prove $(1 - \cos x)/(1 + \cos x) = \tan^2(x/2)$.

2

$$\frac{1 - (2\cos^2(x/2) - 1)}{1 + (2\cos^2(x/2) - 1)} = \frac{2\sin^2(x/2)}{2\cos^2(x/2)} = \tan^2(x/2).$$

The denominators must be nonzero.

Exercise 7F – Pure Y2 P187-188 — Solution 46

3a. Show $\tan\theta + \cot\theta = 2\operatorname{cosec} 2\theta$.

b. Hence find $\tan 75^\circ + \cot 75^\circ$.

3a $\dfrac{\sin\theta}{\cos\theta} + \dfrac{\cos\theta}{\sin\theta} = \dfrac{1}{\sin\theta\cos\theta} = \dfrac{2}{\sin 2\theta} = 2\operatorname{cosec} 2\theta.$

3b

$2\operatorname{cosec} 150^\circ = 2/(1/2) = 4.$

6. Show $\cos^4\theta = 3/8 + (1/2)\cos 2\theta + (1/8)\cos 4\theta$, with full working.

6

$$\cos^4\theta = \left(\frac{1 + \cos 2\theta}{2}\right)^2$$
$$= \frac{1}{4} + \frac{1}{2}\cos 2\theta + \frac{1}{4}\cos^2 2\theta$$
$$= \frac{1}{4} + \frac{1}{2}\cos 2\theta + \frac{1}{8}(1 + \cos 4\theta)$$
$$= \frac{3}{8} + \frac{1}{2}\cos 2\theta + \frac{1}{8}\cos 4\theta.$$

Exercise 7F – Pure Y2 P187-188 — Solution 47

7. Prove $\sin^2(x+y) - \sin^2(x-y) = \sin 2x \sin 2y$.

7

Let $S = \sin x \cos y$, $T = \cos x \sin y$. Addition formulae give $\sin(x+y) = S + T$, $\sin(x-y) = S - T$.

Factor the original difference of squares:

$[(S+T)+(S-T)][(S+T)-(S-T)]$

$= (2S)(2T) = 4 \sin x \cos y \cos x \sin y$

$= (2 \sin x \cos x)(2 \sin y \cos y) = \sin 2x \sin 2y$.

8. Prove $\cos 2\theta - \sqrt{3} \sin 2\theta = 2\cos(2\theta + \pi/3)$.

8

Write

$R\cos(2\theta + \alpha) = R\cos 2\theta \cos\alpha - R \sin 2\theta \sin\alpha$.

Comparing coefficients: $R\cos\alpha = 1$, $R\sin\alpha = \sqrt{3}$.

Divide: $\tan\alpha = \sqrt{3}$, hence $\alpha = \pi/3$. Square and add: $R^2 = 1 + 3 = 4$, so $R = 2$.

$\cos 2\theta - \sqrt{3}\sin 2\theta = 2\cos(2\theta + \pi/3)$.

Exercise 7F – Pure Y2 P187-188 — Solution 48

9. Prove

$4\cos(2\theta - \pi/6)$

$= 2\sqrt{3} - 4\sqrt{3}\sin^2\theta + 4\sin\theta\cos\theta$.

9

$4\cos 2\theta \cos(\pi/6) + 4\sin 2\theta \sin(\pi/6)$

$= 2\sqrt{3}\cos 2\theta + 2\sin 2\theta$

$= 2\sqrt{3}(1 - 2\sin^2\theta) + 4\sin\theta\cos\theta$.

10. Show (a) $\cos\theta + \sin\theta = \sqrt{2}\sin(\theta + \pi/4)$, (b) $\sqrt{3}\sin 2\theta - \cos 2\theta = 2\sin(2\theta - \pi/6)$.

10a $\sqrt{2}[\sin\theta/\sqrt{2} + \cos\theta/\sqrt{2}] = \sin\theta + \cos\theta$.

10b

$2[\sin 2\theta(\sqrt{3}/2) - \cos 2\theta(1/2)]$

$= \sqrt{3}\sin 2\theta - \cos 2\theta$.

Exercise 7F – Pure Y2 P187-188

Solution 49

STEP I 2005 Q4.

a. Given $\cos\theta = 3/5$, $3\pi/2 \le \theta \le 2\pi$, show $\sin 2\theta = -24/25$ and evaluate $\cos 3\theta$.

b. Prove $\tan 3\theta = (3\tan\theta - \tan^3\theta)/(1 - 3\tan^2\theta)$. Hence find $\tan\theta$ if $\tan 3\theta = 11/2$ and $\pi/4 \le \theta \le \pi/2$.

a

$\sin\theta = -4/5$, so $\sin 2\theta = 2(-4/5)(3/5) = -24/25$.

Also $\cos 2\theta = 9/25 - 16/25 = -7/25$.

$\cos 3\theta = \cos 2\theta\cos\theta - \sin 2\theta\sin\theta$

$= (-7/25)(3/5) - (-24/25)(-4/5) = -117/125$.

b

Put $t = \tan\theta$. Using $\tan(2\theta + \theta)$:

$$\tan 3\theta = \frac{2t/(1-t^2) + t}{1 - 2t^2/(1-t^2)} = \frac{3t - t^3}{1 - 3t^2}.$$

The triple-angle identity extends wherever both final sides are defined.

$2t^3 - 33t^2 - 6t + 11 = 0$

$(2t - 1)(t^2 - 16t - 11) = 0$.

Roots: $1/2, 8 \pm \sqrt{75}$. The interval requires $t \ge 1$ and excludes $\theta = \pi/2$ from the given equation, so $t = 8 + \sqrt{75}$.

Very Challenging Exam Example

Solution 50

Edexcel C3 June 2015 Q8.

a. Prove $\sec 2A + \tan 2A = \frac{\cos A + \sin A}{\cos A - \sin A}$, $A \ne (2n+1)\pi/4$, $n \in \mathbb{Z}$.

Tip: the denominator needs a $\cos A - \sin A$ factor, so use $\cos 2A = \cos^2 A - \sin^2 A$.

Another strategy is to work backwards from the desired result, multiplying by $(\cos A + \sin A)/(\cos A + \sin A)$. This suggests the squared numerator. Present reversible steps and check the factors are nonzero.

b. Solve $\sec 2\theta + \tan 2\theta = 1/2$, $0 \le \theta < 2\pi$, to 3 d.p.

a

$$\sec 2A + \tan 2A = \frac{1 + \sin 2A}{\cos 2A}$$

$$= \frac{\cos^2 A + 2\sin A\cos A + \sin^2 A}{(\cos A - \sin A)(\cos A + \sin A)}$$

$$= \frac{(\cos A + \sin A)^2}{(\cos A - \sin A)(\cos A + \sin A)} = \frac{\cos A + \sin A}{\cos A - \sin A}.$$

b

$$\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta} = \frac{1}{2}$$

$2\cos\theta + 2\sin\theta = \cos\theta - \sin\theta \Rightarrow \tan\theta = -1/3$.

$\theta = 2.820, 5.961$.

Both satisfy the original domain.

Modelling

Solution 51

June 2013 (withdrawn), Q8.

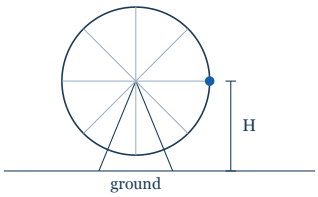
a. Express $9\cos\theta - 2\sin\theta = R\cos(\theta + \alpha)$, $R > 0, 0 < \alpha < \pi/2$, with exact R, α to 4 d.p.

b. State the maximum and the value of θ, $0 < \theta < 2\pi$, where it occurs.

a $R\cos\alpha = 9, R\sin\alpha = 2$. Thus $R = \sqrt{85}$, $\alpha = \tan^{-1}(2/9) = 0.2187$ (4 d.p.).

b(i) Maximum $\sqrt{85}$, since cosine is at most 1.

b(ii) $\theta + \alpha = 2\pi$, so $\theta = 2\pi - \alpha = 6.06$ radians (2 d.p.).



Ruth models a Ferris-wheel passenger height by $H = 10 - 9\cos(\pi t/5) + 2\sin(\pi t/5)$, metres, with t minutes after starting.

c. Find the maximum height and the first time it occurs (2 d.p.).

d. Find the time for two revolutions.

c $H = 10 - \sqrt{85}\cos(\pi t/5 + \alpha)$.
Maximum $10 + \sqrt{85} = 19.22$ m, when $\pi t/5 + \alpha = \pi$.
$t = 5(\pi - \alpha)/\pi = 4.65$ minutes.

d One revolution takes $2\pi/(\pi/5) = 10$ minutes, so two take 20 minutes.

Sine and cosine model oscillations such as tides, pendulums and sound waves.

Test Your Understanding

Solution 52

June 2010 Q7.

a. Express $2\sin\theta - 1.5\cos\theta = R\sin(\theta - \alpha)$, $R > 0, 0 < \alpha < \pi/2$, with α to 4 d.p.

b. Find the maximum and where it occurs for $0 \le \theta < \pi$.

a $R\cos\alpha = 2, R\sin\alpha = 1.5$, so $R = 2.5$, $\alpha = \tan^{-1}(3/4) = 0.6435$.

b(i)
Maximum 2.5, since sine is at most 1.

b(ii)
$\theta - \alpha = \pi/2$, so $\theta = \pi/2 + \alpha = 2.2143\ldots$ radians, in $[0, \pi)$.

Tom models sea height by $H = 6 + 2\sin(4\pi t/25) - 1.5\cos(4\pi t/25)$, $0 \le t < 12$, with t hours after midday.

c. Find the maximum height and when (t to 2 d.p.).

d. Find the times, to the nearest minute, when $H =$ 7 m.

c
$H = 6 + 2.5\sin(4\pi t/25 - \alpha)$.
Maximum 8.5 m, at $t = 25(\pi/2 + \alpha)/(4\pi) = 4.41$ hours after midday.

d
$\sin(4\pi t/25 - \alpha) = 0.4$.
Let $\beta = \arcsin 0.4$. Within the time interval:
$t = \dfrac{25}{4\pi}(\alpha + \beta), \quad \dfrac{25}{4\pi}(\alpha + \pi - \beta)$.
Thus the times are 14:06 and 18:43.

Tip: substitute $\theta = 4\pi t/25$ when using part (a).

6a. Write $65\cos\theta - 20\sin\theta = R\cos(\theta+\alpha)$, $R>0, 0<\alpha<\pi/2$, with α to 4 d.p.

A tourist height on a large wheel is $H = 70 - 65\cos 0.2t + 20\sin 0.2t$, metres; t is minutes after boarding, angles in radians.

b. Find the maximum height.

c. Find the time for one revolution.

d. Find the time spent above 100 m per revolution.

a

$R\cos\alpha = 65, R\sin\alpha = 20.$

Thus $R = \sqrt{4625} = 5\sqrt{185}$,

$\alpha = \tan^{-1}(4/13) = 0.2985.$

$H = 70 - R\cos(0.2t+\alpha).$

b Maximum $70 + 5\sqrt{185} = 138.0$ m (1 d.p.).

c Period $2\pi/0.2 = 10\pi = 31.4$ minutes (1 d.p.).

d

$H > 100 \iff \cos(0.2t+\alpha) < -30/R.$

Let $\beta = \arccos(-30/R)$. The interval above 100 m in one cycle has angular length $2\pi - 2\beta$.

Duration $= \dfrac{2\pi - 2\beta}{0.2} = 11.1$ minutes (1 d.p.).

The first crossing times are about 8.646 and 19.785 minutes.